Discrete Mathematics for Computer Science

B.Tech. (CSE) - First Year, Semester II

Dr. Rishikant Rajdeepak

Assistant Professor
School of Computer Science Engineering & Applications
D Y Patil International University, Akurdi, Pune

AY 2025-26

Course Overview

Course Details

Course Name: Discrete Mathematics for Computer Science

Course Code: CSE1101

Credits: 5

Prerequisites: Intermediate level mathematics

L-T-P-C: 3-1-2-5

Course Objectives

1. To provide a strong foundation in mathematical logic, set theory, and relations for analyzing complex problems.
2. To introduce the concepts of graph theory and algebraic structures as tools for computer science applications.
3. To develop the ability to construct formal mathematical proofs using induction and logical reasoning.
4. To apply discrete structures to represent and solve real-world computational problems.

Course Syllabus — Module I

Module I: Logic and Proof Techniques

Propositions; Logical connectives; Truth tables; Conditional statements, converse, contrapositive, and inverse; Logical equivalence; Tautology; Contradiction; Contingency; Conjunctive normal form; Disjunctive normal form; Translating english sentences; Predicate logic; Universal quantifier; Existential quantifier; Methods of proof; Mathematical induction.

Course Syllabus — Module II

Module II: Sets, Functions, and Relations

Sets; Venn diagrams; Set operations; Subsets; Set identities; Cartesian product of sets; Power set; Cardinality of sets; Basic counting techniques; Principle of inclusion-exclusion; Functions; Injection, surjection, and bijection; Composition of functions; Invertible functions; Relations; Properties of relations; Equivalence relations; Equivalence classes and partitions; Posets; Lattices.

Course Syllabus — Module III

Module III: Graph Theory

Graphs; Types of graphs; Basic terminology in graphs; Handshaking lemma; Paths and cycles; Trees; Complete graphs; Regular graphs; Bipartite and k-partite graphs; Graph representations; Connected graphs; Components of graphs; Euler and Hamiltonian paths; Graph coloring; Chromatic number.

Course Syllabus — Module IV

Module IV: Algebraic structures

Semigroups; Monoids; Groups; Abelian groups; Subgroups; Cyclic groups; Group homomorphism and isomorphism; Rings; Fields;

Laboratory

1. Generate truth tables for simple and compound propositions.
2. Determine logical equivalence among the given propositions using truth tables.
3. Perform set operations on finite sets.
4. Compute the Cartesian product of finite sets and determine its cardinality.
5. Determine whether a given relation is an equivalence relation.
6. Construct a graph from the given edge list and obtain its adjacency matrix.
7. Verify the Handshaking Lemma for a given graph.
8. Construct the Cayley table for a finite set under a given binary operation.

Course Outcomes

At the end of the course the student will be able to:

- Apply rules of propositional logic, predicate logic, and mathematical induction to verify the validity of mathematical arguments and formal proofs.
- Solve problems involving sets, functions, counting techniques, and relations to interpret partitions and equivalence structures.
- Examine graphs to analyze properties such as connectivity, paths, trees, and coloring for structural modeling.
- Implement algebraic structures, including groups, rings, and fields, in suitable mathematical and computational situations.

Textbooks

1. K. H. Rosen, *Discrete Mathematics and its Applications*, 7th ed. McGraw-Hill, 2012.
2. Swapan K. Sarkar, *Textbook of Discrete Mathematics*, 1st ed., 2015.
3. B. Kolman, R. C. Busby, and S. C. Ross, *Discrete Mathematical Structures*, 6th ed., Pearson, 2008.
4. C. L. Liu and D. P. Mohapatra, *Elements of Discrete Mathematics: A Computer Oriented Approach*, 4th ed., Tata McGraw-Hill, 2012.

References

1. NPTEL, "Discrete Mathematics for CS," Prof. Nitin Saxena, IIT Kanpur.
2. MIT OpenCourseWare, "Mathematics for Computer Science," 6.042J.
3. Coursera, "Introduction to Discrete Mathematics for Computer Science Specialization."

Evaluation Scheme

Theory: 60%			
Continuous Assessment	**Mid-term**	**End-term**	**Total**
25	25	50	100

Note: Continuous Assessment within the course will consist only of four quizzes (one per module), contributing 25% to the continuous assessment for theory.

Practical: 40%				
Continuous Assessment			**End-term Practical**	**Total**
Lab Manual	**Lab Assessment**	**Internal Viva-voce**		
10	10	30	50	100

Special Notes

- Please read the 'Student Examination Handbook (2025–2026)' shared by the COE carefully.
- All lectures, tutorials, and study materials will be available in JUNO ERP.
- Maintain a dedicated notebook for this course. Practice regularly and be prepared to present it whenever asked.
- Office hours: Attendance and discussion notes will be kept on record.
  - Div C: Tuesday 09:00 - 10:00
  - Div D: Monday 09:00 - 10:00
- Office: Room 521.

Questions!

Module I: Logic and Proof Techniques

Topics for This Lecture

Propositions • Logical Connectives • Truth Tables

What is Logic?

- Logic is the study of methods and principles used to distinguish correct reasoning from incorrect reasoning.
- In this course, we focus on **propositional logic**.

Basic ideas

- A **proposition** is a declarative sentence that is either true or false, but not both.
- **Propositional logic (propositional calculus)** is the branch of logic that studies propositions and how they combine using logical connectives.

Examples of Propositions

- A proposition must have a definite truth value: **true or false**.

Examples

- $2 + 3 = 5$
- New Delhi is the capital of India.
- 7 is an even number.
- The sun rises in the east.

Each of these statements is either true or false (but not both).

Not Propositions

- Some sentences do **not** have a definite truth value.

Not propositions

- Close the door.
- What time is it?
- $x + 5 = 12$
- Study harder!

Commands, questions, and statements with unknown variables are **not** propositions.

Propositional Variables and Truth Values

- We use letters to represent propositions:

$$p, q, r, s, \ldots$$

  They are called **propositional variables**.
- Each propositional variable has a **truth value**:

$$\text{True (T)} \quad \text{or} \quad \text{False (F)}$$

Examples

- p: 2 is an even number. (T)
- q: 5 is a prime number. (T)
- r: $3 + 4 = 10$. (F)

Compound Propositions

- Propositions can be combined to form **compound propositions**.
- We use logical connectives to build more complex statements.

Example

p : It is raining. q : I am carrying an umbrella.

$p \wedge q$: It is raining and I am carrying an umbrella.

The truth of a compound proposition depends on the truth values of the propositions that form it.

Logical Connectives

Precedence	Symbol	Meaning
1	$\neg p$	Negation: not p
2	$p \wedge q$	Conjunction: p and q
3	$p \vee q$	Disjunction (inclusive or): p or q (or both)
3	$p \oplus q$	Exclusive or (XOR): exactly one of p, q is true
4	$p \rightarrow q$	Implication: if p, then q
5	$p \leftrightarrow q$	Biconditional: p iff q

Note that OR and XOR have same precedence order. Parentheses can always be used to avoid ambiguity.

Truth Tables

- A **truth table** shows the truth value of a compound proposition for every possible combination of truth values of its variables.

Example

- $\neg p$ means "not p".
- It reverses the truth value of p.

p	$\neg p$
T	F
F	T

Truth Table Example with OR

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Observation:

- $p \vee q$ is false only when both p and q are false.

Truth Table Example with AND

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Observation:

- $p \wedge q$ is true only when both p and q are true.

Truth Table Example with Exclusive OR (XOR)

p	q	$p \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

Observation:

- $p \oplus q$ is true only when either p or q is true (not both).

Implication

- $p \to q$ means "If p, then q".
- It is false only when p is true and q is false.

p	q	$p \to q$
T	T	T
T	F	F
F	T	T
F	F	T

Biconditional

- $p \leftrightarrow q$ means "p if and only if q".

  p is the hypothesis, and q is the conclusion.
- True when p and q have the same truth value.

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

Example

Let:

p: "It is raining"

q: "I carry an umbrella"

- $\neg p$: It is not raining.
- $p \wedge q$: It is raining and I carry an umbrella.
- $p \vee q$: It is raining or I carry an umbrella (or both).
- $p \oplus q$: Either it is raining or I carry an umbrella, but not both.
- $p \rightarrow q$: If it is raining, then I carry an umbrella.
- $p \leftrightarrow q$: I carry an umbrella if and only if it is raining.

Exercises — Propositions

- Decide whether each of the following is a proposition. If it is, give its truth value.
  1. $2 + 5 = 9$
  2. Close the window.
  3. Every prime number is odd.
  4. $x + 3 = 10$
  5. Paris is the capital of France.

Exercises — Negation

- Write the negation of each statement in simple English.
  1. The class starts at 9:00.
  2. The student submitted the assignment.
  3. The number 7 is even.
  4. All computers in the lab are working.

Exercises — Conjunction and Disjunction

- Express each sentence using symbols and then find its truth value.
  1. It is raining and it is cold.
  2. She is a teacher or she is a doctor.
  3. The bus is late and the road is empty.
  4. He will pass the exam or he will repeat the course.

Exercises — Exclusive OR

- Rewrite each sentence using exclusive or ($\oplus$).
  1. You can choose tea or coffee, but not both.
  2. The room will be locked by Alice or Bob, but not both.
  3. The light is on in exactly one of the two rooms.

Exercises — Implication

- Translate each into symbolic form.
  1. If it rains, the match will be cancelled.
  2. If a number is even, then it is divisible by 2.
  3. If the student studies, then the student passes.

Exercises — Biconditional

- Write each as a biconditional statement.
  1. A figure is a square if and only if it has four equal sides and four right angles.
  2. You may enter the lab if and only if you have your ID card.
  3. A number is even if and only if it is divisible by 2.

Module I: Logic and Proof Techniques

Topics for This Lecture

Conditional statements • Converse • Contrapositive • Inverse

Conditional Statements

- A conditional statement has the form

$$p \rightarrow q$$

- Read as: **If p, then q.**
- p is the hypothesis (antecedent)
- q is the conclusion (consequent)

Example

If a number is divisible by 4, then it is even.

Truth of a Conditional

- $p \to q$ is false only when

$$p \text{ is true and } q \text{ is false.}$$

- In every other case, it is true.

p	q	$p \to q$
T	T	T
T	F	F
F	T	T
F	F	T

Converse, Inverse, Contrapositive

Given the conditional

$$p \to q$$

- **Converse:** $q \to p$
- **Inverse:** $\neg p \to \neg q$
- **Contrapositive:** $\neg q \to \neg p$

Important

$$p \to q \quad \text{and} \quad \neg q \to \neg p$$

have the same truth value.

Example — Converse, Inverse, Contrapositive

Original statement

If it is raining, then the ground is wet.

- **Converse:** If the ground is wet, then it is raining.
- **Inverse:** If it is not raining, then the ground is not wet.
- **Contrapositive:** If the ground is not wet, then it is not raining.

Logical Equivalence

- Two compound propositions are **logically equivalent** if they have the same truth value in every possible case.

$$p \equiv q \quad \text{means "}p \text{ and } q \text{ are logically equivalent."}$$

Example

$$\neg(p \wedge q) \equiv \neg p \vee \neg q$$

(This is one of De Morgan's laws.)

Truth Table: Conditional, Converse, Inverse, Contrapositive

p	q	$p \to q$	$q \to p$	$\neg p \to \neg q$	$\neg q \to \neg p$
T	T	T	T	T	T
T	F	F	T	T	F
F	T	T	F	F	T
F	F	T	T	T	T

Observation:

- $p \to q \equiv \neg q \to \neg p$.
- $q \to p \equiv \neg p \to \neg q$.

Logical Equivalence Using Truth Tables

$$p \to q \equiv \neg p \vee q$$

p	q	$p \to q$	$\neg p \vee q$
T	T	T	T
T	F	F	F
F	T	T	T
F	F	T	T

Both columns are identical, so the statements are logically equivalent.

Module I: Logic and Proof Techniques

Topics for This Lecture

Logical Equivalence • Tautology • Contradiction • Contigency

Tautology, Contradiction, and Contingency

- A **tautology**: always true.
- A **contradiction**: always false.
- A **contingency**: sometimes true, sometimes false.

$$p \vee \neg p \quad \text{(tautology)}$$

$$p \wedge \neg p \quad \text{(contradiction)}$$

Logical Equivalence

Two propositions are **logically equivalent** if they always have the same truth value.

$$p \equiv q$$

This symbol "$\equiv$" is NOT a logical connective. It means: "these two statements have identical truth tables."

Another way to say the same thing:

$$(p \leftrightarrow q) \text{ is a tautology.}$$

Example

$$p \rightarrow q \equiv \neg p \vee q$$

Logical Equivalences — I

- Identity laws:

$$p \wedge T \equiv p \qquad p \vee F \equiv p$$

- Domination laws:

$$p \vee T \equiv T \qquad p \wedge F \equiv F$$

- Idempotent laws:

$$p \vee p \equiv p \qquad p \wedge p \equiv p$$

- Double negation laws:

$$\neg(\neg p) \equiv p$$

Logical Equivalences — II

- Commutative laws:

$$p \vee q \equiv q \vee p \qquad p \wedge q \equiv q \wedge p$$

- Associative laws:

$$(p \vee q) \vee r \equiv p \vee (q \vee r)$$

$$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$$

Logical Equivalences — III

- Distributive laws:

$$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$$

$$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$$

- De Morgan laws:

$$\neg(p \wedge q) \equiv \neg p \vee \neg q$$

$$\neg(p \vee q) \equiv \neg p \wedge \neg q$$

Logical Equivalences — IV

- Absorption laws:

$$p \vee (p \wedge q) \equiv p$$

$$p \wedge (p \vee q) \equiv p$$

- Negation laws:

$$p \vee \neg p \equiv T$$

$$p \wedge \neg p \equiv F$$

Equivalences Involving Conditionals

$$p \to q \equiv \neg p \vee q$$

$$p \to q \equiv \neg q \to \neg p$$

$$\neg(p \to q) \equiv p \wedge \neg q$$

$$(p \to q) \wedge (p \to r) \equiv p \to (q \wedge r)$$

$$(p \to r) \wedge (q \to r) \equiv (p \vee q) \to r$$

Equivalences Involving Biconditionals

$$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$$

$$p \leftrightarrow q \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$$

$$\neg(p \leftrightarrow q) \equiv p \leftrightarrow \neg q$$

Exercises — Logical Equivalence

Do not use truth tables. Use logical equivalence laws only.

- Show that
$$\neg(p \to q) \quad \text{and} \quad p \wedge \neg q$$
are logically equivalent.
- Show that
$$\neg(p \vee (\neg p \wedge q)) \quad \text{and} \quad \neg p \wedge \neg q$$
are logically equivalent by developing a series of logical equivalences.
- Show that
$$(p \wedge q) \to (p \vee q)$$
is a tautology.

Building Blocks

- A **literal** is a propositional variable or its negation:

$$p, \ \neg p$$

- A **conjunction** is an AND of literals:

$$(p \wedge q \wedge \neg r)$$

- A **disjunction** is an OR of literals:

$$(p \vee \neg q \vee r)$$

- A single conjunction or disjunction may contain **one or many literals**.

Disjunctive Normal Form (DNF)

A formula is in **disjunctive normal form** if it is a disjunction of one or more conjunctions of literals.

General form:

$$(p \wedge q \wedge r) \ \vee \ (\neg p \wedge q) \ \vee \ \cdots$$

Each conjunction represents one complete way for the formula to be true.

Examples of DNF

- $(p \wedge q) \vee (\neg r \wedge s)$
- $(p \wedge q \wedge \neg r) \vee (s \wedge t)$
- $p \vee (q \wedge r)$

Each expression is an OR of one or more AND-clauses.

Conjunctive Normal Form (CNF)

A formula is in **conjunctive normal form** if it is a conjunction of one or more disjunctions of literals.

General form:

$$(p \vee q \vee r) \ \wedge \ (\neg p \vee r) \ \wedge \ \cdots$$

Each disjunction eliminates one way the formula could be false.

Examples of CNF

- $(p \vee q) \wedge (\neg r \vee s)$
- $(p \vee \neg q \vee r) \wedge (\neg p \vee r)$
- $p \wedge (q \vee r \vee s)$

Each expression is an AND of one or more OR-clauses.

Example: Converting to DNF

Convert the following into DNF:

$$(p \vee q) \wedge (r \vee s)$$

Apply distributive laws:

$$(p \wedge r) \vee (q \wedge r) \vee (p \wedge s) \vee (q \wedge s)$$

The result is a disjunction of conjunctions of literals.

Example: Converting to CNF

Convert the following into CNF:

$$p \vee (q \wedge r)$$

Apply distributive laws:

$$(p \vee q) \wedge (p \vee r)$$

The result is a conjunction of disjunctions of literals.

Exercises — Identification

Classify each expression as **CNF, DNF, both, or neither**.

(1) $(p \wedge q \wedge r) \vee (\neg p \wedge s)$

(2) $(p \vee q \vee r) \wedge (\neg p \vee s) \wedge t$

(3) $(p \wedge q) \vee (r \vee s)$

(4) $\neg(p \vee q)$

Exercises — Conversion

Rewrite each expression as indicated.

(1) Convert to DNF:

$$p \wedge (q \rightarrow r)$$

(2) Convert to CNF:

$$(p \wedge q) \vee (r \wedge s)$$

(3) Rewrite in both CNF and DNF:

$$(p \vee q) \rightarrow r$$

Module I: Logic and Proof Techniques

Topics for This Lecture

Predicates • Quantifiers

Predicate Logic

- Propositional logic cannot describe statements involving variables.
- **Predicate logic** extends propositional logic by introducing:
  - variables
  - predicates
  - quantifiers

A **predicate** is a statement containing variables, such as:

$$P(x) : \text{“}x \text{ is even”}$$

Predicates and Domains

- A predicate becomes a proposition when a value is assigned to its variable.
- The set from which variables take values is called the **domain**.

Examples

- $P(x) : x > 3$, domain: integers
- $Q(x, y) : x + y = 10$, domain: real numbers

Universal Quantifier

- The **universal quantifier** is denoted by $\forall$.
- $\forall x\, P(x)$ means:

"For all x, $P(x)$ is true."

Example

$$\forall x(x^2 \geq 0)$$

Universal Quantifier — Examples

- $\forall x(x + 1 > x)$, domain: real numbers
- $\forall n(n^2 \geq n)$, domain: integers
- $\forall x(x$ is a student$)$ — depends on domain

The truth of $\forall x\, P(x)$ depends on the domain.

Existential Quantifier

- The **existential quantifier** is denoted by $\exists$.
- $\exists x\, P(x)$ means:

"There exists an x such that $P(x)$ is true."

Example

$$\exists x(x^2 = 4)$$

Existential Quantifier — Examples

- $\exists x(x^2 = 2)$, domain: real numbers
- $\exists n(n$ is prime and even$)$
- $\exists x(x < 0)$

Translating English Sentences

- English sentences often use words like:
  - all, every
  - some, there exists
  - none
- These are translated using quantifiers.

Translation Examples

Let the domain be integers.

English: Every integer is even or odd.

$$\forall x(x \text{ is even } \vee x \text{ is odd})$$

English: There exists an integer that is even and prime.

$$\exists x(x \text{ is even } \wedge x \text{ is prime})$$

More Translation Examples

Let the domain be students.

English: Every student has an ID card.

$$\forall x(x \text{ is a student } \rightarrow x \text{ has an ID card})$$

English: Some student has not submitted the assignment.

$$\exists x(x \text{ is a student } \wedge \neg S(x))$$

Exercises — Predicate Logic and Quantifiers

Translate into logical expressions.

(1) Every real number has a square greater than or equal to zero.

(2) There exists an integer that is divisible by both 2 and 3.

(3) Every student in the class has submitted the assignment.

(4) Some student in the class has not attended the lecture.

State the domain clearly in each case.

Exercises — Truth and Meaning

(1) Determine whether $\forall x(x^2 > x)$ is true when the domain is:

- integers
- real numbers

(2) Determine whether $\exists x(x^2 = x)$ is true for integers.

(3) Give an example where $\forall x\, P(x)$ is false but $\exists x\, P(x)$ is true.

Module I: Logic and Proof Techniques

Topics for This Lecture

Methods of Proofs • Mathematical Induction

Methods of Proof

In mathematics, statements are established using logical arguments called **proofs**.

Common methods of proof include:

- Direct proof
- Proof by contrapositive
- Proof by contradiction
- Proof by cases
- Disproof by counterexample
- Mathematical induction

Direct Proof

A **direct proof** assumes the hypothesis is true and uses logical steps to show that the conclusion follows.

Typical form:

$$p \to q$$

Assume p is true and prove q.

Example

If n is an even integer, then n^2 is even.

Direct Proof — Example

Assume n is even.

$$n = 2k \quad \text{for some integer } k$$

Then

$$n^2 = (2k)^2 = 4k^2 = 2(2k^2)$$

Hence, n^2 is even.

Proof by Contrapositive

To prove

$$p \to q$$

we instead prove the logically equivalent statement:

$$\neg q \to \neg p$$

This method is useful when the contrapositive is easier to prove.

Example
If n^2 is odd, then n is odd.

Proof by Contrapositive — Example

We prove the contrapositive:
If n is even, then n^2 is even.

Assume $n = 2k$.
Then

$$n^2 = 4k^2$$

So n^2 is even.
Therefore, if n^2 is odd, n must be odd.

Proof by Contradiction

To prove a statement p, assume $\neg p$ is true and derive a contradiction.

A contradiction means a statement of the form:

$$r \wedge \neg r$$

Example

There is no smallest positive rational number.

Proof by Contradiction — Example

Assume there exists a smallest positive rational number r.
Then

$$\frac{r}{2}$$

is also a positive rational number and

$$\frac{r}{2} < r$$

This contradicts the assumption that r is the smallest.
Hence, no smallest positive rational number exists.

Proof by Cases

A statement is proved by dividing the argument into a finite number of cases and proving each case.

Example

For every integer n, $n^2 \geq n$.

Proof by Cases — Example

Case 1: $n \geq 1$

$$n^2 \geq n$$

Case 2: $n = 0$

$$n^2 = n = 0$$

Case 3: $n \leq 0$

$$n^2 \geq 0 \geq n$$

Thus, $n^2 \geq n$ for all integers n.

Disproof by Counterexample

A universal statement can be disproved by finding a single example for which it is false.

Example

Statement: All prime numbers are odd.

Counterexample:

$$2 \text{ is prime and even}$$

Mathematical Induction

Mathematical induction is used to prove statements of the form:

$$P(n) \text{ is true for all integers } n \geq n_0$$

It consists of two steps:

- Base step
- Inductive step

Principle of Mathematical Induction

- **Base step:** Show $P(n_0)$ is true.
- **Inductive hypothesis:** Assume $P(k)$ is true.
- **Inductive step:** Show $P(k+1)$ is true.

If both steps hold, then $P(n)$ is true for all $n \geq n_0$.

Induction — Example

Prove that

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}$$

for all integers $n \geq 1$.

Induction — Proof

Base step ($n = 1$):

$$1 = \frac{1(2)}{2}$$

Inductive hypothesis: Assume true for $n = k$.

$$1 + 2 + \cdots + k = \frac{k(k+1)}{2}$$

Inductive step:

$$1 + 2 + \cdots + k + (k+1) = \frac{k(k+1)}{2} + (k+1)$$

$$= \frac{(k+1)(k+2)}{2}$$

Exercises — Methods of Proof

(1) Prove directly that the sum of two even integers is even.

(2) Prove by contrapositive: If n^2 is divisible by 3, then n is divisible by 3.

(3) Prove by contradiction that $\sqrt{2}$ is irrational.

(4) Disprove the statement: For all integers n, $n^2 + n + 41$ is prime.

(5) Prove by cases that $|x| \geq 0$ for all real numbers x.

Exercises — Mathematical Induction

(1) Prove using induction that

$$1 + 3 + 5 + \cdots + (2n - 1) = n^2$$

for all $n \geq 1$.

(2) Prove using induction that

$$2^n \geq n + 1$$

for all integers $n \geq 1$.

(3) Prove that

$$3 \mid (n^3 - n)$$

for all integers $n \geq 1$.

Module I: Logic and Proof Techniques

Tutorial

Exercises — Propositions

(1) Decide whether each sentence is a proposition. Justify your answer.

- The sum of two even integers is even.
- Read the instructions carefully.
- $n^2 - 1 = 0$.
- There are infinitely many prime numbers.

(2) Give two different propositions that have the same truth value.

Exercises — Logical Connectives

Let p: "The server is running", q: "The network is available", r: "The website is accessible".

(1) Write the following statements using symbols:

- The server is running and the network is available.
- The website is accessible only if the server is running.
- Either the network is available or the website is accessible, but not both.

(2) Write an English sentence corresponding to:

$$(p \wedge q) \rightarrow r$$

Exercises — Truth Tables

(1) Construct the truth table for:

$$(p \vee q) \wedge \neg r$$

(2) Construct the truth table for:

$$\neg(p \wedge q) \rightarrow r$$

(3) From the truth table, determine whether

$$(p \vee q) \rightarrow (q \vee p)$$

is a tautology.

Exercises — Conditional Statements

(1) Write the converse, inverse, and contrapositive of:

If a function is differentiable, then it is continuous.

(2) Identify which of the four statements (original, converse, inverse, contrapositive) are logically equivalent.

Exercises — Logical Equivalence

Do not use truth tables.

(1) Show that

$$p \vee (q \wedge p) \equiv p$$

(2) Show that

$$(p \to r) \wedge (q \to r) \equiv (p \vee q) \to r$$

(3) Determine whether

$$(p \leftrightarrow q) \wedge (q \leftrightarrow r)$$

is logically equivalent to

$$p \leftrightarrow r$$

Exercises — Logical Equivalence

Do not use truth tables. Use logical equivalence laws only.

(1) Show that

$$\neg(p \wedge (p \vee q)) \equiv \neg p$$

(2) Show that

$$(p \to q) \wedge (p \to r) \equiv p \to (q \wedge r)$$

(3) Show that

$$(p \vee q) \wedge (\neg p \vee r) \equiv (p \to r) \vee q$$

(4) Determine whether

$$(p \leftrightarrow q) \vee (p \leftrightarrow \neg q)$$

is a tautology.

(5) Show that

$$\neg(p \vee q) \vee (p \wedge q) \equiv \neg p \leftrightarrow \neg q$$

(6) Determine whether

$$(p \to q) \leftrightarrow (\neg q \to \neg p) \text{ is a tautology.}$$

Exercises — Tautology, Contradiction, Contingency

(1) Classify each proposition as a tautology, contradiction, or contingency:

- $(p \wedge q) \rightarrow p$
- $(p \wedge \neg p) \vee q$
- $p \leftrightarrow \neg p$

(2) Give a proposition involving two variables that is neither a tautology nor a contradiction.

Exercises — CNF and DNF

Rewrite each proposition in the required normal form.

(1) Rewrite the following in **disjunctive normal form**:

$$(p \vee q) \wedge (r \vee s)$$

(2) Rewrite the following in **conjunctive normal form**:

$$p \vee (q \wedge r)$$

(3) Rewrite the following in **disjunctive normal form**:

$$\neg(p \to q)$$

Exercises — CNF and DNF

(4) Rewrite the following in **conjunctive normal form**:

$$(p \wedge q) \vee r$$

(5) Rewrite the following in **both CNF and DNF**:

$$(p \vee q) \rightarrow r$$

(6) Classify the following expression as **CNF, DNF, both, or neither**:

$$(p \vee q) \wedge (\neg p \vee r)$$

Exercises — Translating English Sentences

Translate into logical notation. State the domain clearly.

(1) Every rational number has a decimal expansion.

(2) There exists a natural number that is divisible by every natural number.

(3) No employee in the company works on all days of the week.

(4) There is a machine that can perform every assigned task.

Exercises — Predicate Logic and Quantifiers

(1) Let $P(x)$: "x is divisible by 3". Write in symbols:

- Every integer greater than 6 is divisible by 3.
- Some integer greater than 6 is divisible by 3.

(2) Write the negation of:

$$\exists x \, \forall y \, (x \leq y)$$

(3) Determine whether the statement below is true or false. Justify your answer.

$$\forall x \, \exists y \, (x + y = 0)$$

Exercises — Direct Proof

(1) Prove directly that the sum of two odd integers is even.

(2) Prove that if n is an integer divisible by 6, then n is divisible by 3.

(3) Prove that the product of two consecutive integers is even.

Exercises — Proof by Contrapositive

(1) Prove by contrapositive: If n^2 is even, then n is even.

(2) Prove by contrapositive: If ab is irrational, then at least one of a or b is irrational.

(3) Prove by contrapositive: If n is not divisible by 3, then n^2 is not divisible by 3.

Exercises — Proof by Contradiction

(1) Prove by contradiction that $\sqrt{3}$ is irrational.

(2) Prove that there is no smallest positive real number.

(3) Prove that there are infinitely many prime numbers.

Exercises — Proof by Cases

(1) Prove by cases that for every integer n,

$$n^2 \geq n$$

(2) Prove that $|x| = x$ if $x \geq 0$ and $|x| = -x$ if $x < 0$.

(3) Prove by cases that n^2 is either even or odd for every integer n.

Exercises — Disproof by Counterexample

(1) Disprove the statement: All integers of the form $n^2 + n + 41$ are prime.

(2) Disprove: For all real numbers x, $\sqrt{x^2} = x$.

(3) Disprove: Every continuous function is differentiable.

Exercises — Mathematical Induction

(1) Prove using induction that

$$1 + 2 + 4 + \cdots + 2^n = 2^{n+1} - 1$$

for all integers $n \geq 0$.

(2) Prove using induction that

$$n! \geq 2^{n-1}$$

for all integers $n \geq 1$.

(3) Prove using induction that

$$1^2 + 2^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

for all integers $n \geq 1$.

Thank You!